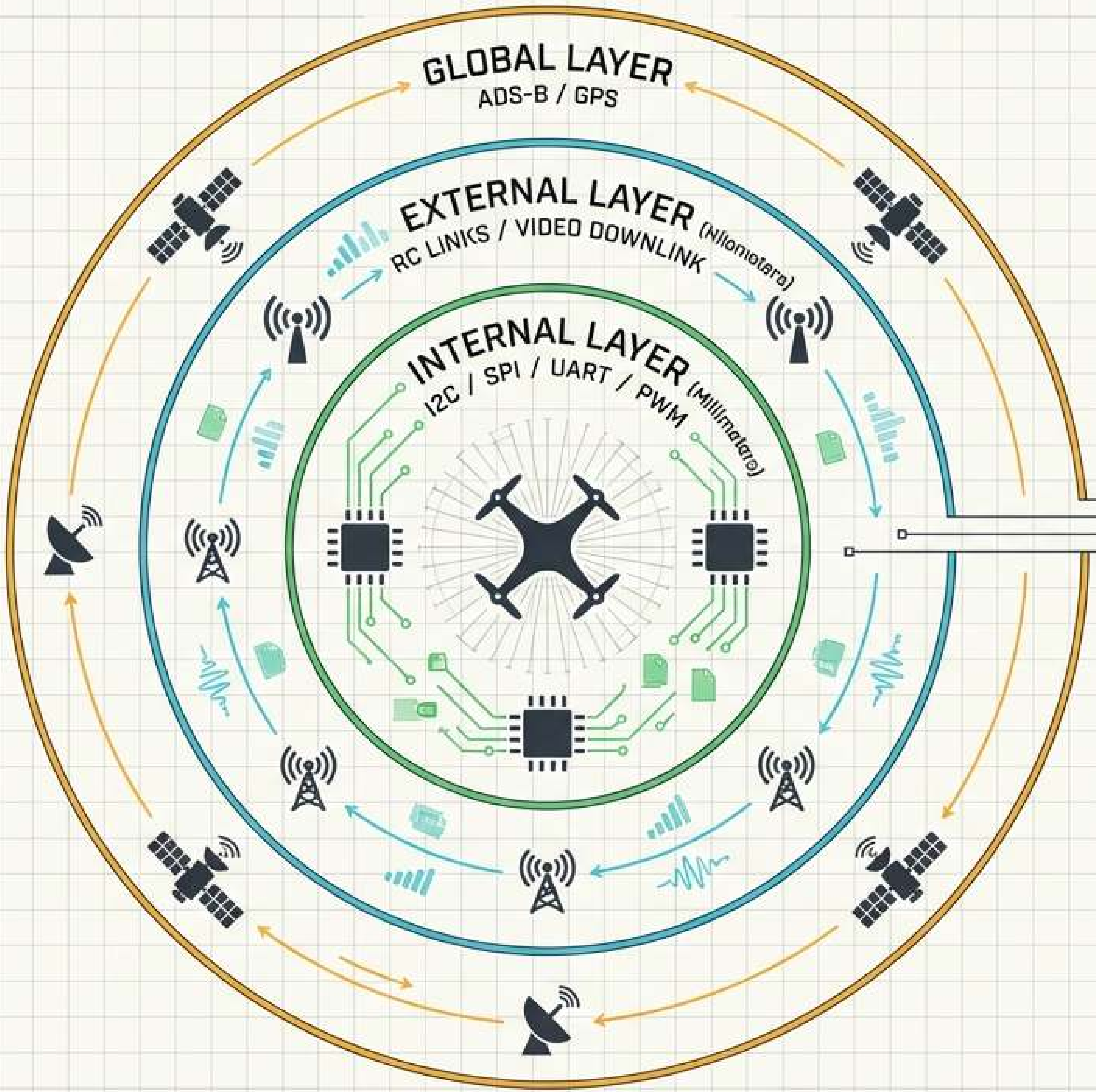
The Architecture of UAV Data Transmission

Deconstructing the Communication Ecosystem from Radio Frequencies to Micro-Buses

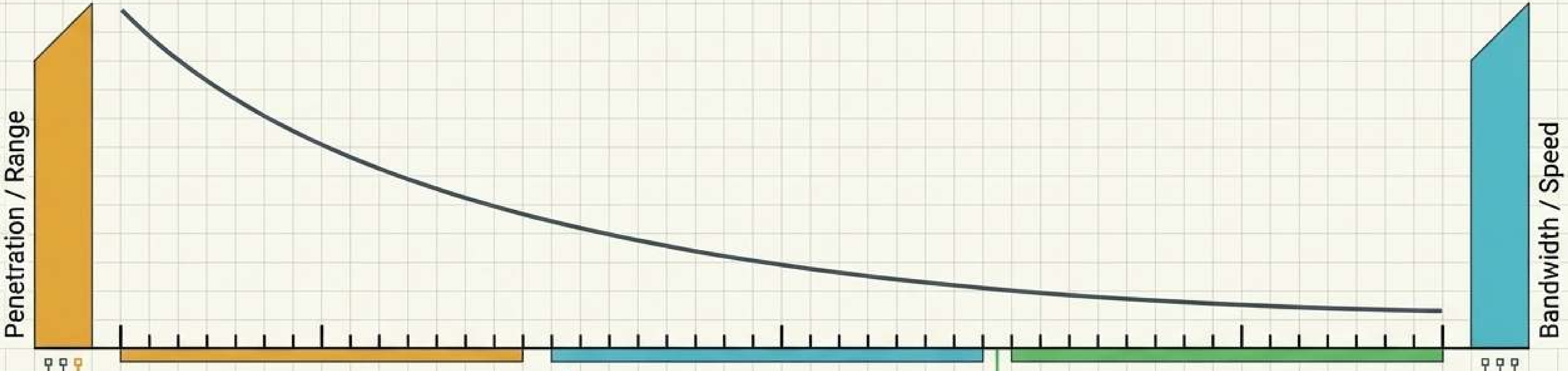


THREE LAYERS OF UAV CONNECTIVITY

A drone relies on a highly synchronized stack of communication networks operating across vastly different physical scales—from extraterrestrial satellite pings to millimeter-scale circuit traces.

The Physics of the Radio Spectrum

UAV communication operates on a strict physics trade-off between range and data rate.



900MHz (868/915MHz)

Maximum object
Maximum object penetration.
Historically the choice for long-range flying.

2.4GHz

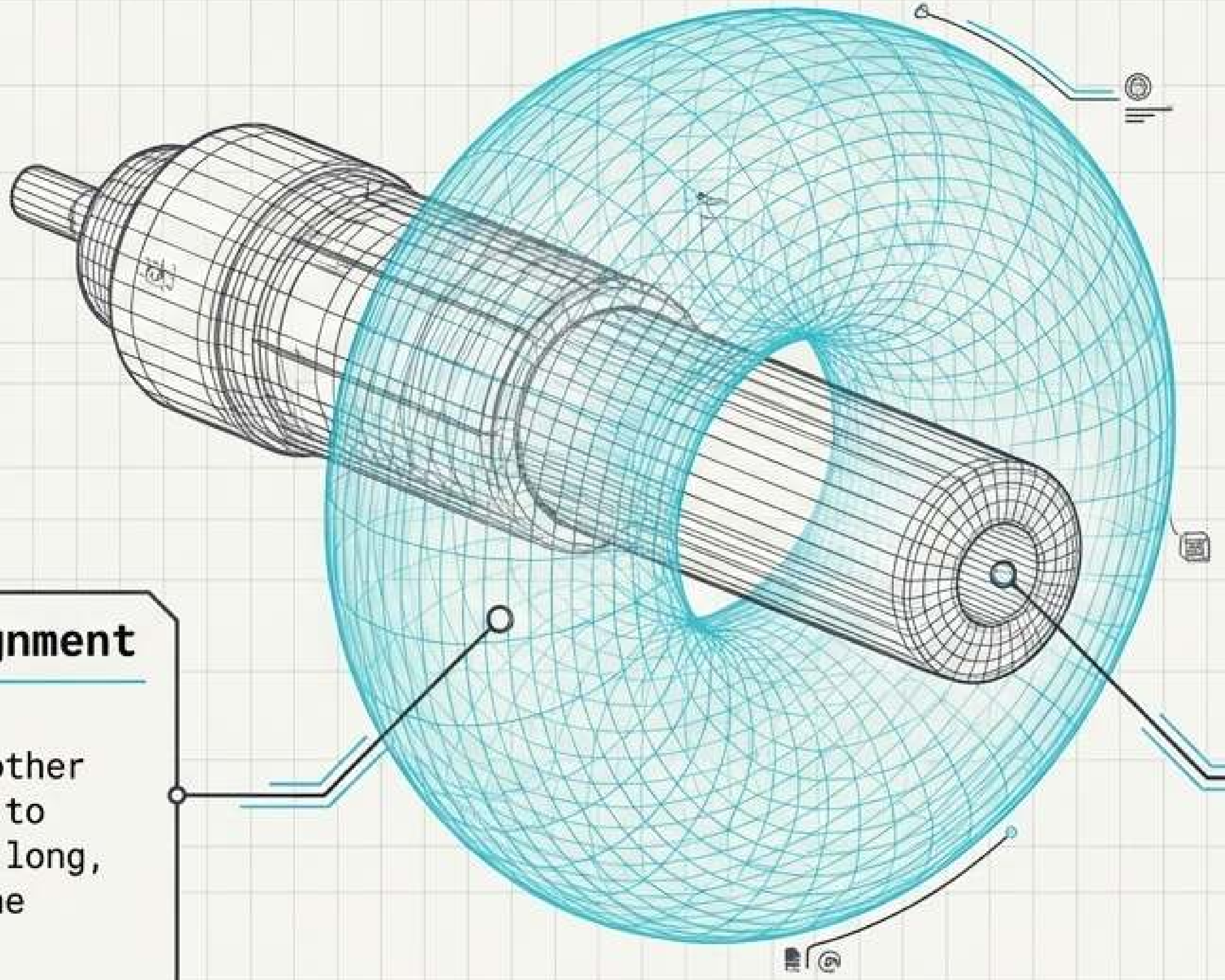
The modern RC standard.
Fast packet rates.
Can achieve 60+ miles with LoRa modulation.

5.8GHz

High bandwidth, low penetration.
The global standard for dense analog and digital video transmission.

Optimal Transmission Zones and Antenna Alignment

Radio waves do not fire out of an antenna like a laser.
They radiate outward in a torus (donut) shape.



Broadside Alignment

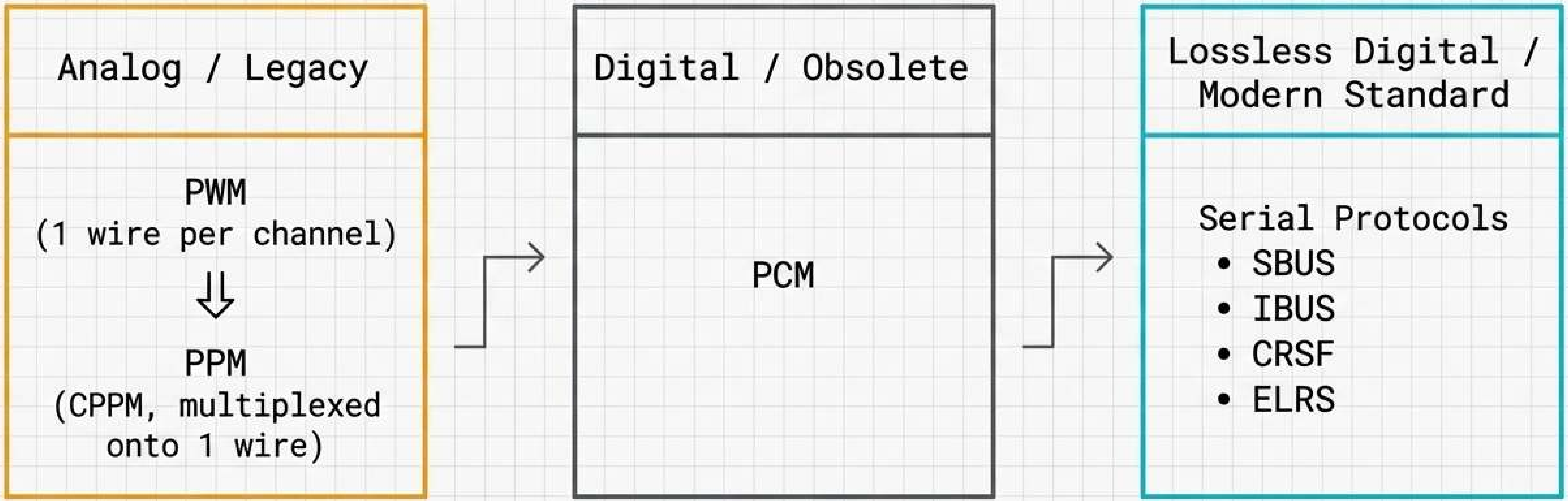
Position antennas parallel to each other and perpendicular to the aircraft. The long, flat side emits the strongest signal.

The Dead Zone

Pointing the tip of the antenna directly at the aircraft pushes it into the weakest part of the radiation field, causing sudden signal drops.

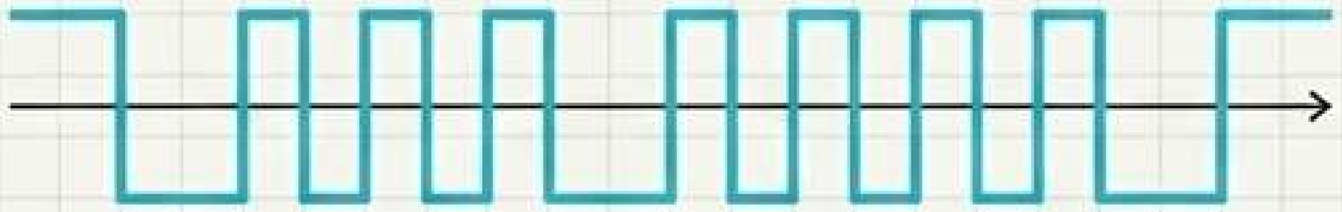
The Evolution of Control Links

The industry has completely transitioned from bulky, interference-prone analog setups to streamlined, lossless digital serial protocols requiring only 3 wires (Signal, Power, Ground).



Serial Protocols: SBUS vs IBUS

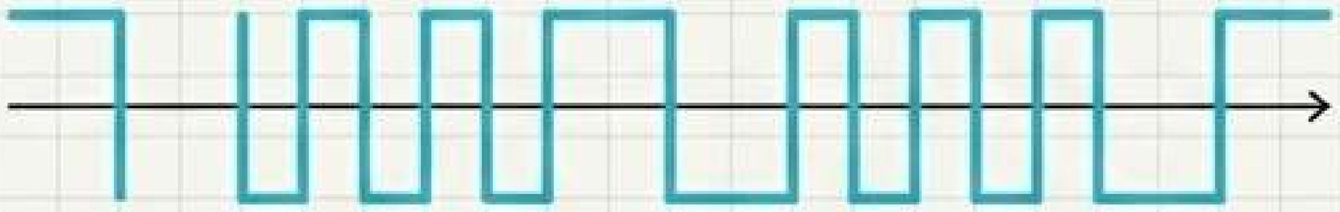
IBUS (FlySky)



A standard, non-inverted UART protocol (115200 baud, 8 data bits, no parity, 1 stop bit).

Highly documented, bidirectional (RC + telemetry), and natively decoded by virtually any MCU.

SBUS (Futaba/FrSky)



Encodes up to 18 PWM channels into a single signal. Crucially, it utilizes an inverted UART logic.

Older flight controllers (like the Naze32) required dedicated hardware inverters to decode it.

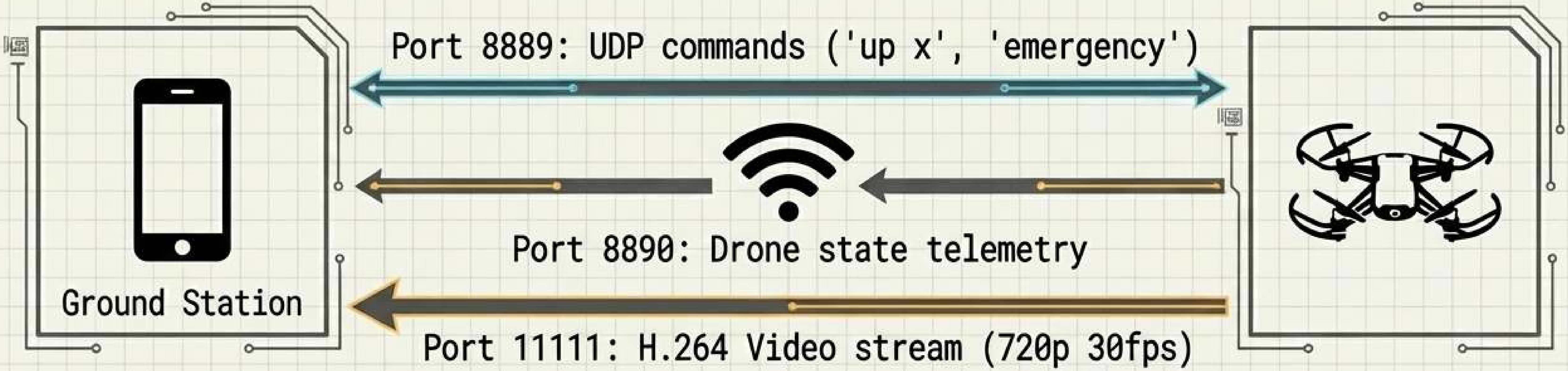
The Heavyweight Matrix: ExpressLRS vs. TBS Crossfire

Feature	ExpressLRS (ELRS)	TBS Crossfire (CRSF)
Nature	Open-Source	Proprietary
Frequencies	2.4GHz & 900MHz	868/915MHz
Max Packet Rate	1000Hz (2.4GHz) / 200Hz (900MHz)	150Hz
Modulation	LoRa at all packet rates	LoRa at 50Hz, drops to FSK at 150Hz
Security	Unencrypted	Encrypted
Hardware	SX127x with ESP8285/ESP32	Closed Ecosystem

Verdict: ELRS dominates modern builds due to LoRa penetration at higher frequencies and unmatched 1000Hz low-latency performance.

IP-Based Macro-Links: The UDP Architecture

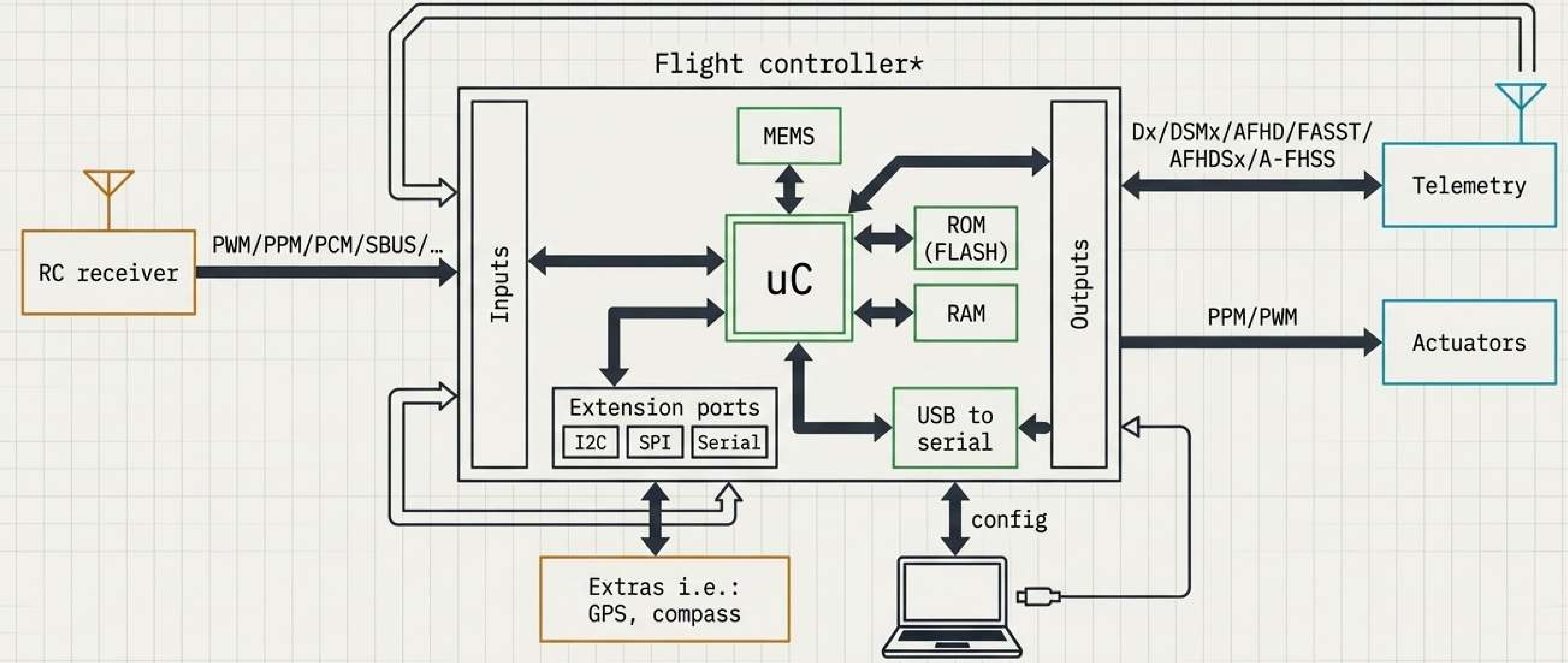
Commercial 'toy' drones (like DJI Tello) operate as flying WiFi access points (192.168.10.1). Instead of RC protocols, they use standard IP networking.



Trade-off: Ubiquitously compatible with mobile devices, but suffers from high latency, dropped ISO/OSI stacks, and crowded 2.4GHz urban interference.

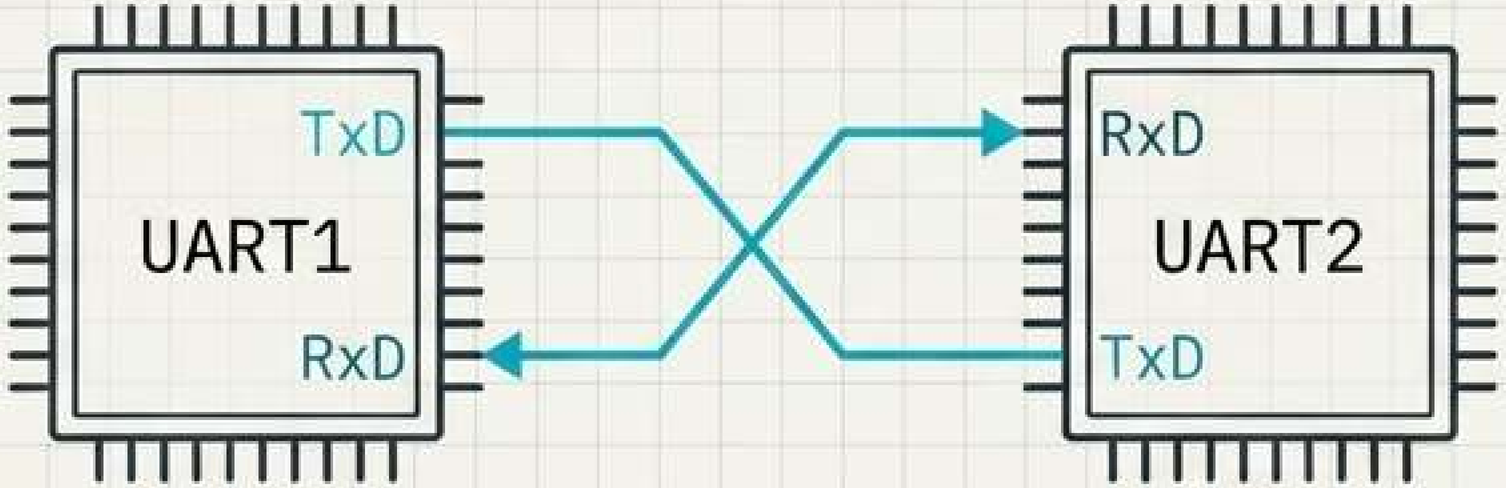
The Micro-Scale Hub: Inside the Flight Controller

The Flight Controller (FC) is the central switchboard. It receives external macro-signals (RC, GPS), processes them via an MCU (STM/ARM), and routes commands through micro-scale digital buses (I2C, SPI) to actuators and telemetry downlinks.



Internal Digital Protocol: UART

Universal Asynchronous Receiver-Transmitter



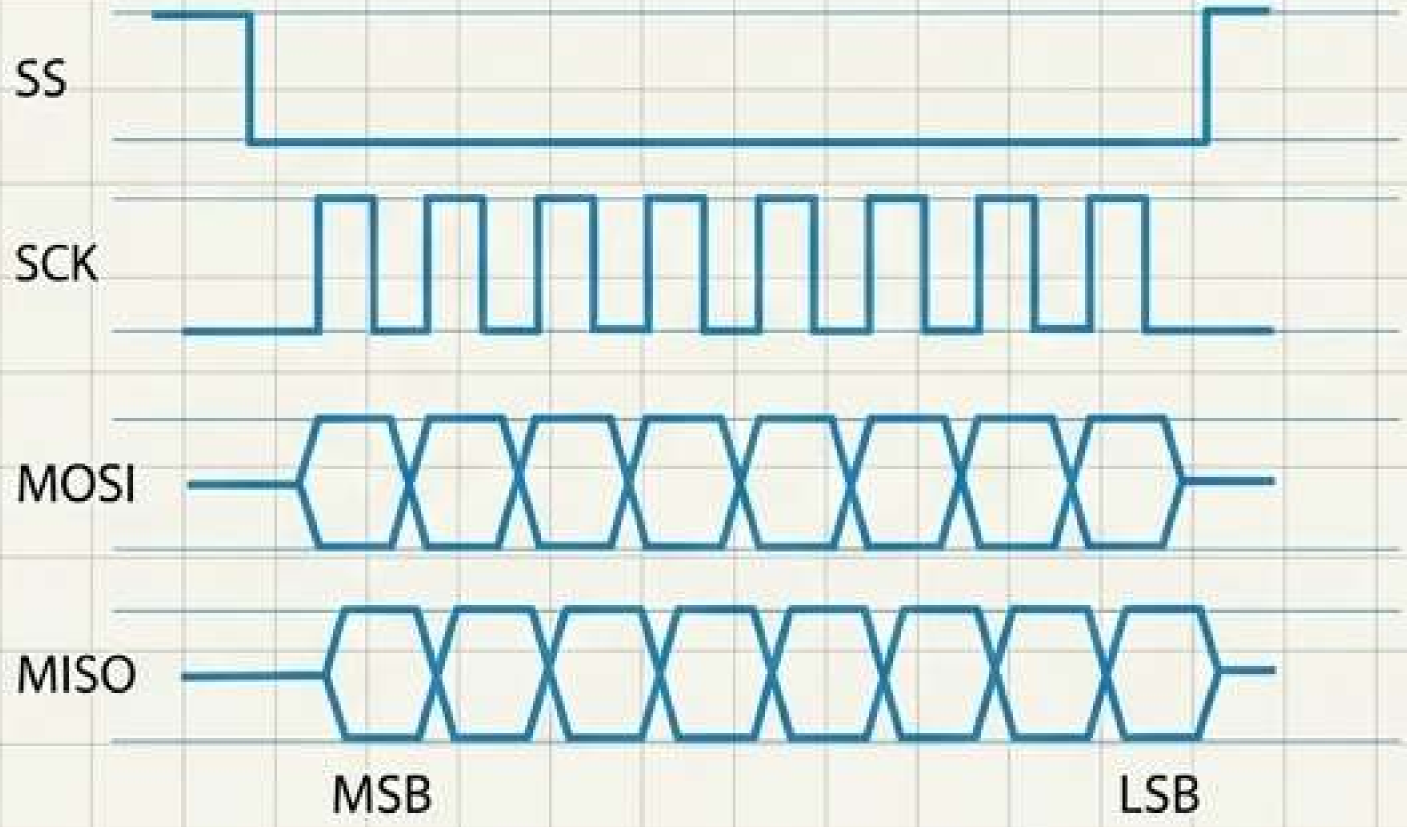
TECHNICAL SPECIFICATIONS

- **Topology:** 1-to-1.
- **Timing:** Asynchronous (no shared clock signal).
- **Wiring:** 2 dedicated data wires. The Transmit (TxD) of one chip connects to the Receive (RxD) of the other.
- **Primary UAV Uses:** Flashing the FC, telemetry routing, and communicating with external systems (like GPS or serial receivers).

Internal Digital Protocol: SPI

Serial Peripheral Interface

TIMING DIAGRAM



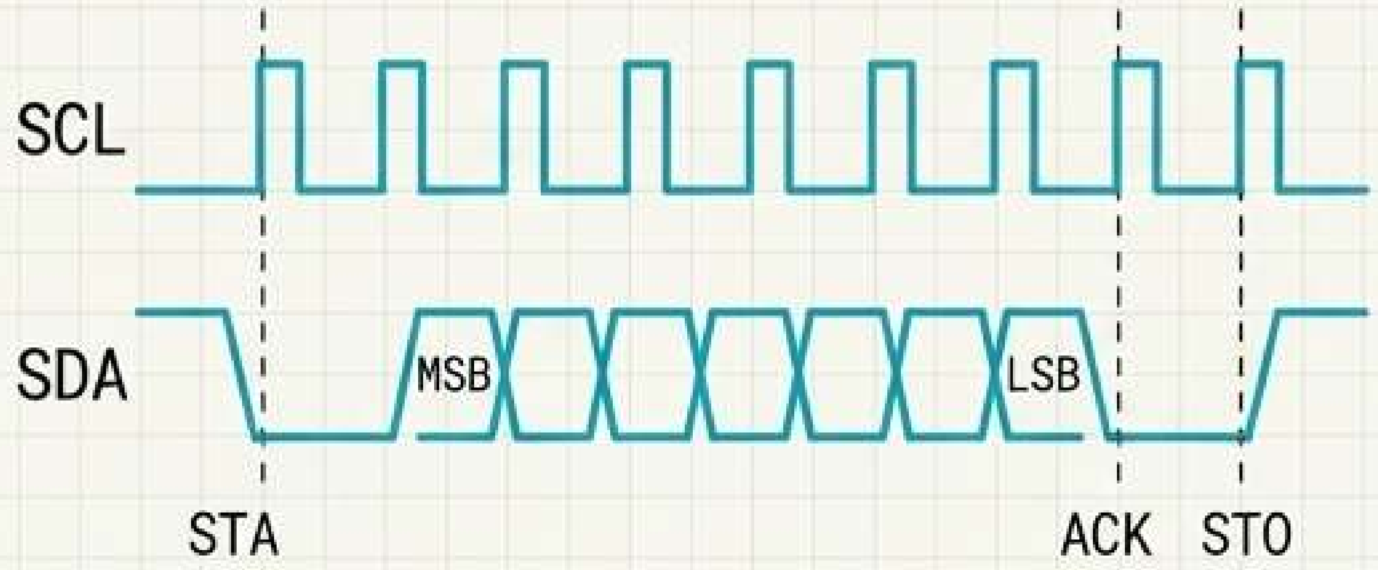
TECHNICAL SPECIFICATIONS

- **Topology:** 1 Master to multiple Slaves.
- **Timing:** Synchronous (relies on a shared clock line).
- **Wiring:** 4-wire bus (MOSI, MISO, SCK, SS).
- **Configuration:** The mode must precisely match between Master and Slave. The most common configuration in UAVs is Mode 0 (CPOL=0, CPHA=0).
- **Purpose:** Built for extremely high-speed, on-board data transfer.

Internal Digital Protocol: I2C (TWI)

Inter-Integrated Circuit

TIMING DIAGRAM

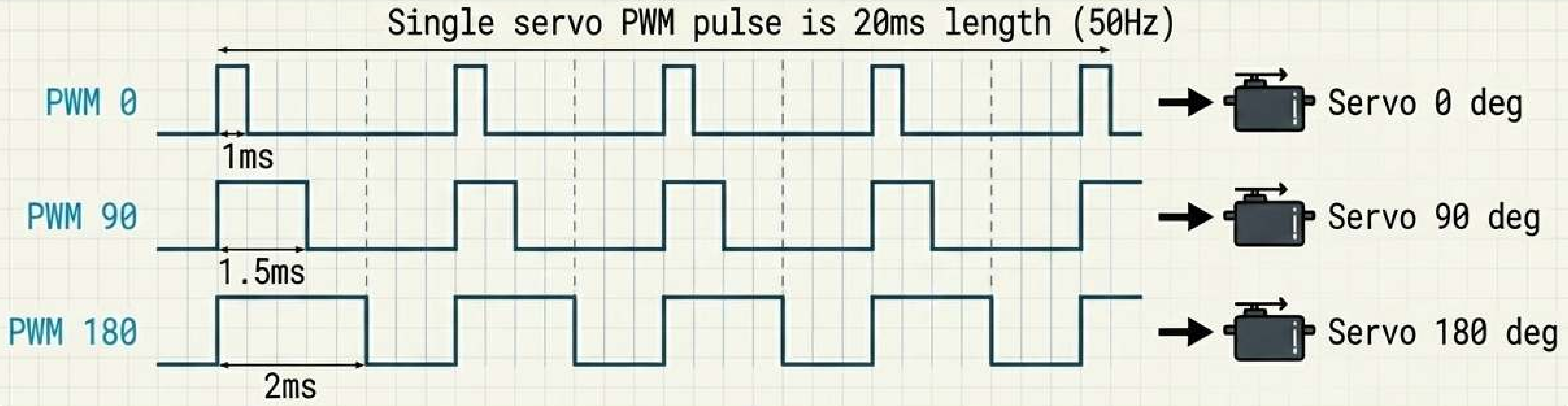


TECHNICAL SPECIFICATIONS

- **Topology:** 1-to-many.
- **Timing:** Synchronous.
- **Wiring:** Highly efficient 2-wire bus (SDA for data, SCL for clock).
- **Primary UAV Uses:** Because it only requires two wires, I2C is the dominant protocol for wiring internal drone sensors like Barometers, IMUs, Compasses (magnetometers), and proximity sensors using 3.3Vr 5V logic.

The Actuator Link: PWM

Internal communication ends at the **ESC** (Electronic Speed Controller) or servo, driven by **Pulse Width Modulation (PWM)**. The ratio of active signal to off-signal (duty cycle) dictates exact motor angle or speed.



- **Standard Analog:** Operates at 50Hz (20ms period).
- **Digital Actuators:** Push frequencies up to 300Hz for lower latency.

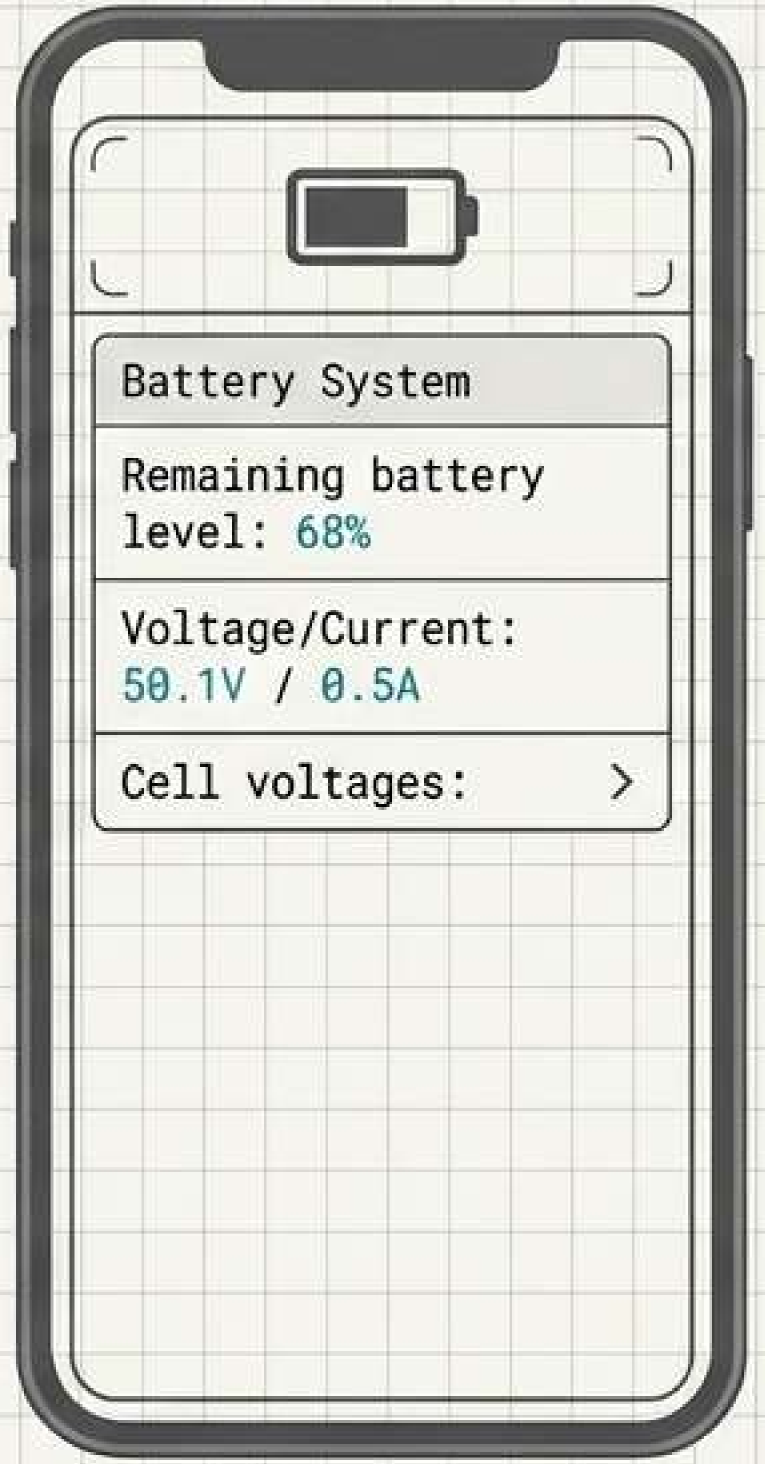
Duty Cycle Mapping: 1ms = 0° (Throttle Off) | 1.5ms = 90° (Mid) | 2ms = 180° (Max Throttle)

The Telemetry Downlink

Modern protocols (like FrSky, ELRS, and IBUS) are inherently bidirectional. The drone continuously feeds an OSD (On-Screen Display) or radio screen.

Key Data Streams:

- **VBAT**: Real-time battery voltage and current draw.
- **RSSI & LQ**: Received Signal Strength Indicator and Link Quality warn pilots before a failsafe drop.
- **Addressing**: Protocols like IBUS use sensor addressing (up to 16 devices) with address 0 strictly reserved for voltage monitoring.



The Video Downlink: Analog vs. Digital

Analog (The Racer's Choice)

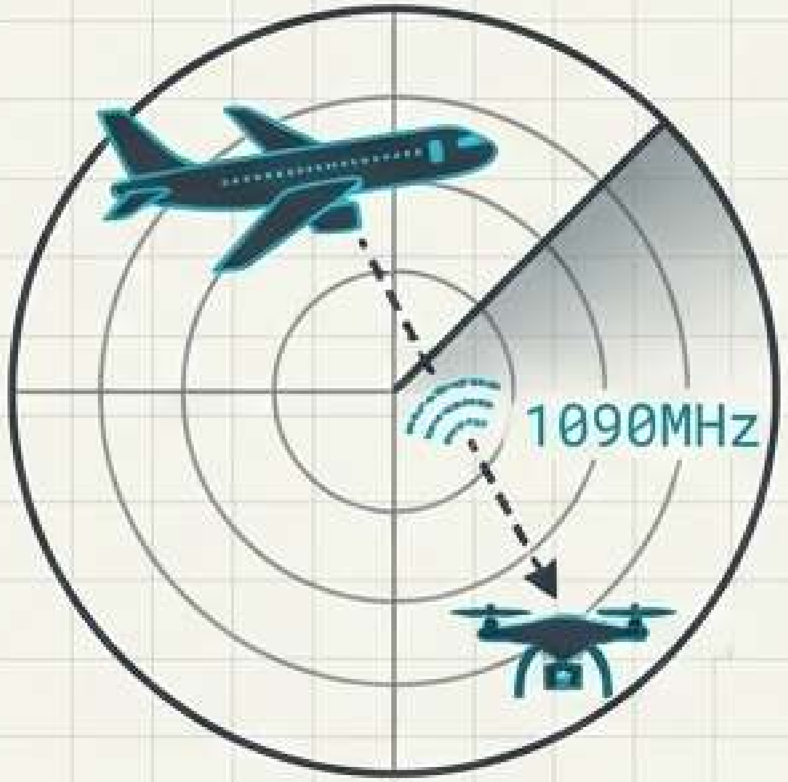
Uses 5.8GHz transmitting NTSC/PAL standard lines. Max power capped by regulation (often 200mW). Unmatched for zero-latency racing, but degrades via gradual static and interference.

Digital HD (The Cinematographer's Choice)

Transmits compressed MPEG streams (H.264/H.265) at 720p/1080p. Delivers a flawless image but introduces digital processing latency (up to 1 second) and suffers from harsh packet loss freezes rather than static.

Global Awareness: The ADS-B Network

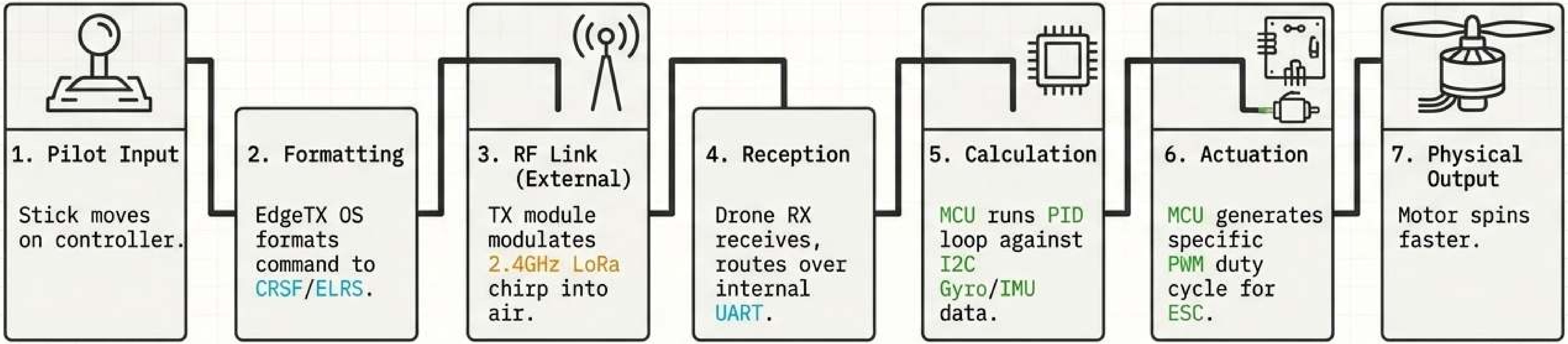
At the highest layer of the communication ecosystem is ADS-B (Automatic Dependent Surveillance-Broadcast).



- Operating on a digital 1090MHz unencrypted protocol, commercial planes broadcast their ID, GPS position, heading, speed, and altitude every second.
- Modern UAVs integrate ADS-B receivers (like the Orange Cube) to automatically detect manned aircraft.
- Enables execution of TCAS-style collision avoidance maneuvers, representing the absolute macro-level of drone communication.

Synthesis: The Complete Signal Chain

How a finger movement becomes physical flight in less than 5 milliseconds:



The 2025 Modern UAV Standard

While the landscape of acronyms is vast, the modern standard has firmly consolidated around a specific, highly optimized stack:

Blueprint Checklist

- **Firmware:** EdgeTX
  Open source, highly configurable.
- **Control Link:** ExpressLRS 2.4GHz
  1000Hz packet rate, LoRa modulation.
- **Internal Routing:** UART & I2C
  Non-inverted UART for telemetry/control; I2C for peripheral sensors.
- **Video:** 5.8GHz
  Digital HD (for commercial/cinema) or Analog (for ultra-low latency racing).